EXPERIMENT-06

SIMULATION-BASED INVESTIGATION OF MICROWAVE SEMICONDUCTOR DEVICES AND FREQUENCY-CONVERSION CHARACTERISTICS OF A MICROWAVE MIXER

PROGRAM

```
clc;

clear;

close all;

% Simulation-Based Investigation of Microwave Semiconductor Devices

% and Frequency-Conversion Characteristics of a Microwave Mixer

%% 1. Parameters

fRF = 2.4e9;

fLO = 2.3e9;

fs = 20e9;

T = 100e-9;

t = 0:1/fs:T-1/fs;

VRF = 0.5;

VLO = 1.0;

R = 50;

%% 2. Semiconductor Diode I-V Characteristics

Is = 1e-12;

n = 1.5;

Vt = 0.02585;

Vd = linspace(-0.5,0.8,500);

Id = Is*(exp(Vd/(n*Vt))-1);

Id = min(Id,1e3);
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```
%% 3. RF and Local Oscillator Signals

RF = VRF*cos(2*pi*fRF*t);

LO = VLO*cos(2*pi*fLO*t);

%% 4. Mixer Frequency Conversion

vMixer = RF.*LO;

IF = (VRF*VLO/2)*cos(2*pi*(fRF-fLO)*t);

fIF = abs(fRF-fLO);

fSum = fRF+fLO;

%% 5. Frequency Spectrum

N = length(t);

Y = abs(fft(vMixer))/N;

Y = Y(1:floor(N/2)+1);

Y(2:end-1) = 2*Y(2:end-1);

faxis = (0:floor(N/2))*fs/N;

%% 6. Display Results

fprintf('\nMICROWAVE MIXER SIMULATION RESULTS\n');

fprintf('-----------------------------------\n');

fprintf('RF Frequency = %.2f GHz\n',fRF/1e9);

fprintf('LO Frequency = %.2f GHz\n',fLO/1e9);

fprintf('IF Frequency = %.2f MHz\n',fIF/1e6);

fprintf('Sum Frequency = %.2f GHz\n',fSum/1e9);

fprintf('Load Resistance = %.2f Ohms\n',R);

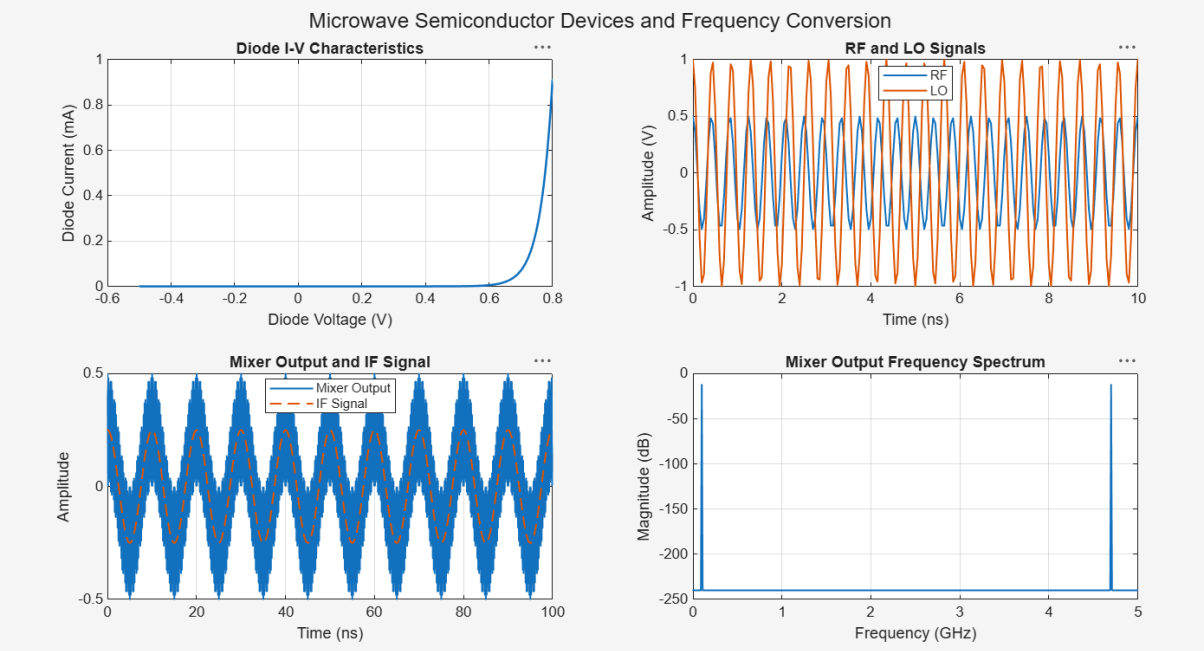
%% 7. Combined Subplot Figure

figure('Name','Microwave Semiconductor and Mixer',...
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```
    'NumberTitle','off');
% Subplot 1: Diode I-V Characteristics
subplot(2,2,1);
plot(Vd,Id*1e3,'LineWidth',1.5);
grid on;
xlabel('Diode Voltage (V)');
ylabel('Diode Current (mA)');
title('Diode I-V Characteristics');
% Subplot 2: RF and LO Signals
subplot(2,2,2);
plot(t*1e9,RF,'LineWidth',1.2);
hold on;
plot(t*1e9,LO,'LineWidth',1.2);
grid on;
xlim([0 10]);
xlabel('Time (ns)');
ylabel('Amplitude (V)');
title('RF and LO Signals');
legend('RF','LO','Location','best');
% Subplot 3: Mixer Output
subplot(2,2,3);
plot(t*1e9,vMixer,'LineWidth',1.2);
hold on;
plot(t*1e9,IF,'--','LineWidth',1.2);
```

```
grid on;
xlim([0 100]);
xlabel('Time (ns)');
ylabel('Amplitude');
title('Mixer Output and IF Signal');
legend('Mixer Output','IF Signal','Location','best');
% Subplot 4: Frequency Spectrum
subplot(2,2,4);
plot(faxis/1e9,20*log10(Y+1e-12),'LineWidth',1.2);
grid on;
xlim([0 5]);
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('Mixer Output Frequency Spectrum');
sgtitle('Microwave Semiconductor Devices and Frequency Conversion');
```

OUTPUT



RESULT:

The simulation demonstrates diode behavior and microwave frequency conversion. For the input frequencies of 2.4 GHz and 2.3 GHz, the difference frequency is 100 MHz, and the sum frequency is 4.7 GHz.